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Video: Chain Rule

7 min

Quiz: Chain Rule Quiz

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Chain Rule Quiz

Review Learning Objectives

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Due Mar 10, 11:59 PM IST

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1. Find the derivative of $f(x) = (x^3 - 2x + 5)^4$

1 / 1 point

- ☒ $4(x^3 - 2x + 5)^3 \cdot (3x^2 - 2)$
- ☐ $(x^3 - 2x + 5)^3$
- ☐ $(x^3 - 2x + 5)^3 \cdot (3x^2 - 2x)$

Try again

Correct

Let $g(x) = x^3 - 2x + 5$ and $f(x) = g^4$

$$f' = 4g^3 = 4(x^3 - 2x + 5)^3$$

$$g'(x) = 3x^2 - 2$$

Applying chain rule i.e. $f(g(x))' = f'(g(x))g'(x)$, we get,

$$f'(g(x)) = 4(x^3 - 2x + 5)^3$$

$$g' = 3x^2 - 2$$

$$f(g(x))' = f'(g(x))g'(x) = 4(x^3 - 2x + 5)^3 \cdot (3x^2 - 2)$$

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2. For a variety of problems, we will often find that we get an equation including the exponential.

1 / 1 point

Consider the example $f(x) = e^{2x^2}$. We can view this in the context of the chain rule by formulating it as $f(x) = h(g(x))$, where $h(g(x)) = e^{2x^2}$ and $g(x) = 2x^2$. Given that, find the derivative of $f(x) = e^{2x^2}$.

- ☐ $f'(x) = e^{4x}$
- ☒ $f'(x) = 4xe^{2x^2}$
- ☐ $f'(x) = e^{4x+4} \cdot (x + 1)$

Correct

Note the general form of the chain rule for a function $f(x) = h(g(x))$ has a derivative

$$f'(x) = h'(g(x)) \cdot g'(x)$$

Applying the chain rule for this problem, $h(g(x)) = e^{2x^2}$, and $h'(g(x)) = e^{2x^2}$, because recall $\frac{d}{dx}e^x = e^x$. We can see that inside the exponent, $g(x) = 2x^2$, and $g'(x) = 4x$.

